Department of Electrical Engineering (Fall 2025)

ME2002- Applied Thermodynamics
Quiz # 2
(CLO 02)

Roll No: _____
Name: _____

Time: 20 min
Max Marks: 20

Q1

Calculate energy of the following system using energy equations.

A steam turbine receives steam from two boilers. One flow passes through a pipe having cross sectional area 0.1 m² with a velocity of 23.3397m/s at 3.5 MPa, 700°C, just before entering the turbine. Whereas the other flow passes through a pipe having cross sectional area 0.2 m² with a velocity of 33.24825m/s at 800 kPa, 500°C, just before entering the turbine. The exit point is single and exit state is uniform 10 kPa, two phase state. If the output power produced by the turbine is 45184.646 kW, find out the mass flow rate of liquid water at the exit of turbine.

Solution

Data:

$A_1 = 0.1\ m^2$
$V_1 = 23.3397\ m/s$
$P_1 = 3.5\ MPa$

$A_2 = 0.2\ m^2$
$V_2 = 33.24825$
$P_2 = 800\ kPa.$

$\dot{W} = 45184.646\ KW$

Steady-flow energy Eq →

$$\dot{W} = \dot{m}_1 \left(h_1 + \frac{V_1^2}{2000} \right) + \dot{m}_2 \left(h_2 + \frac{V_2^2}{2000} \right) - \dot{m}_e h_e \quad (2)$$

$h_e = h_f + x \cdot h_{fg}$

$$\Rightarrow \dot{m}_1 = \dot{V}_1 = \frac{A_1 V_1}{v_1} \quad (2)$$

$$\dot{m}_2 = \dot{V}_2 = \frac{A_2 V_2}{v_2} \quad (2)$$

Superheated Vapor Water

With interpolation → 3000KP and 4000KPa at 700°C. Table B.13

we can find v_1 and h_1. (2)

$v_1 = 0.11718$
$h_1 = 3922.9$

from Tables → 800KPa at 500°C

$v_2 = 0.44331$ $h_2 = 3480.60$ (1)

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exit $P = 10\ KPa \rightarrow$

From Table B1.2 $\rightarrow$ $h_f = 191.81$ (1) $h_g = 2584.63$.

$h_{fg} = 2392.82$.

$\dot{m}_1 = \frac{A_1 \cdot V}{v}$ (2)

$= 19.917\ kg/s$

$\dot{m}_2 = \frac{A_2 \cdot V}{v}$ (2).

$= 15.074\ kg/s$.

$\dot{m}_c = \dot{m}_1 + \dot{m}_2$

$= 34.991$

$\dot{W} = \dot{m}_1\left(h_1 + \frac{V_1^2}{2000}\right) + \dot{m}_2\left(h_2 + \frac{V_2^2}{2000}\right) - \dot{m}_e h_e$

$h_e = 2441.79$ (2)

$\therefore h_e = h_f + x\, h_{fg}$.

$x = \frac{h_e - h_f}{h_{fg}}$.

$x = 0.94065$ (2)

Mass flow rate.

$\dot{m}_f = \dot{m}_e (1 - x)$.

$\dot{m}_f = 34.991(1 - 0.94065)$ (2)

$\boxed{\dot{m}_f = 2.077\ kg/s.}$

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